

Task: Loan Approval Prediction Pipeline

GitHub Repository: https://github.com/Onlyy-Anand/Loan_analyse

Dataset : <https://www.kaggle.com/datasets/bhanupratapbiswas/loan-approval-prediction-case-study>

Notebook Link: https://colab.research.google.com/drive/1boHlz2Nqn_OZulxvC427iUqoTqE_Ffg8?usp=sharing

Executive Summary

Objective The goal of this project was to build a supervised machine learning pipeline to predict whether a borrower's loan will be approved based on historical applicant features. The primary challenge involved handling data inconsistencies and addressing a class imbalance favoring approved loans.

Methodology The dataset underwent a strict preprocessing pipeline to ensure model robustness:

- **Imputation:** Missing numerical values (like loan amount and term) were filled using the median to avoid outlier skew, while categorical missing values were filled with the most frequent class.
- **Encoding & Scaling:** Categorical variables were one-hot encoded, and numerical features were standardized (mean of 0, standard deviation of 1).
- **Imbalance Handling:** Instead of synthesizing data, the models were initialized with balanced class weights, heavily penalizing the misclassification of the minority class (rejected loans).

Model Evaluation Two primary models were evaluated: Logistic Regression and Random Forest.

- **Logistic Regression** emerged as the superior model, achieving a Precision of 0.872, Recall of 0.882, and an F1-Score of 0.877 on the positive class. Most importantly, it achieved an ROC-AUC of 0.854, demonstrating a strong ability to distinguish between good and bad loans.
- **Random Forest** yielded a higher recall (0.906) but fell short in overall discriminative power with an ROC-AUC of 0.806.

Business Recommendations The Logistic Regression model successfully flags ~71% of bad loans, representing a 31% improvement in default prevention over a baseline, unweighted model.

- **Deployment Threshold:** The statistically optimal decision threshold (Youden's J) is 0.488.
- **Strategic Adjustment:** During economic downturns, the institution should increase this threshold to 0.60+ to prioritize capital preservation. In growth phases, dropping it closer to 0.49 maximizes market share while maintaining a calculated risk profile.

- Screenshot 1:

...		Precision	Recall	F1-Score	ROC-AUC
	Logistic Regression	0.872	0.882	0.877	0.854
•	Random Forest	0.837	0.906	0.870	0.806

- Chart 1:

